

What are conditional statements?

In []: A conditional statements **is** used to make decision **in** a program.

In []: Syntax: **if** condition:
 statement

In []: If executes different blocks of code based on whether condition **is True or False**

1)if statement:

Syntax: if condition: statement

```
In [1]: age=20
        if age>=18:
            print("Eligible for voting")
```

Eligible for voting

```
In [2]: marks=85
        if marks>=35:
            print("Pass")
```

Pass

2)ifelse statement

In []: If the condition **is True**, the **if** block executes;
otherwise, the **else** block executes.

Syntax:

```
In [ ]: if condition:
        statement
        else:
            statement
```

```
In [4]: age=int(input("enter Age:"))
        if age>=18:
            print("Eligible for Voting")
        else:
            print("Not eligible")
```

enter Age: 19

Eligible for Voting

Even or Odd

```
In [6]: num=int(input("Enter a number:"))
        if num%2==0:
            print("Even Number")
        else:
            print("Odd Number")
```

Enter a number: 4

Even Number

Positive and Negative

```
In [8]: num=int(input("Enter number:"))
        if num>=0:
            print("The number is positive")
        else:
            print("The number is negative")
```

Enter number: 7

The number is positive

if....elif...else Statement

```
In [ ]: Multiple condition we are using here.
```

Syntax

```
In [ ]: if condition:
        statement
        elif condition:
            statement
        else:
            statement
```

```
In [13]: marks=int(input("Enter Marks:"))
if marks>=90:
    print("Grade A")
elif marks>=75:
    print("Grade B")
elif marks>=50:
    print("Grade C")
elif marks>=35:
    print("Grade D")
else:
    print("Fail")
```

Enter Marks: 98

Grade A

```
In [16]: temp=int(input("Enter temp:"))
if temp>40:
    print("Very Hot")
elif temp>30:
    print("Hot")
elif temp>20:
    print("Pleasant")
else:
    print("Cold")
```

Enter temp: 19

Cold

Nested if

An if statement inside another if

ATM

```
In [21]: pin=1234
entered_pin=int(input("Enter PIN:"))
if entered_pin==pin:
    balance=5000
    print("Balance:",balance)
    if balance>=1000:
        print("Transaction Successful")
    else:
        print("Transaction Fail")
else:
    print("Invalid PIN")
```

Enter PIN: 1234

Balance: 5000

Transaction Successful

Discount Calculator

```
In [27]: amt=float(input("Enter Purchase Amount:"))
if amt>=5000:
    discount=amt*0.10
    final=amt-discount
    print("Discount:",discount)
    print("Final Amount:",final)
else:
    print("No discount")
```

Enter Purchase Amount: 7000

Discount: 700.0

Final Amount: 6300.0

In []: